MIL-S-1222H
21 October 1986
SUPERSEDING
MIL-S-001222G(SHIPS)
13 January 1976
MIL-S-1222F
9 December 1963
(See 6.3 and 6.7)

MILITARY SPECIFICATION

STUDS, BOLTS, HEX CAP SCREWS, SOCKET HEAD CAP SCREWS AND NUTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standard studs, bolts, hex cap screws, socket head cap screws, and nuts in diameters of 1/4 inch and larger for critical applications where a high degree of reliability is required.

1.2 Classification. Fasteners shall be of the following types and styles, as specified (see 6.2.1):

Studs

Type I - pull body
Type II - Reduced body
Type III - Constant strength body

Style	Stud end thread	Nut end thread
a	class 5	Class 3A
b	class 5	Class 2A
c	Class 3A	Class 3A
d	Class 3A	Class 2A

Type IV - Continuous thread

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Bolts

Type I - Hex
Type II - Heavy hex
Type III - Heavy hex structural
Type IV - Round head
Type V - Round head square neck

Screws

Type I - Hex cap (finished hex bolt)
Type II - Hex socket head cap

Nuts

Type I - Hex
Type II - Hex jam
Type III - Heavy hex
Type IV - Heavy hex jam
Type V - Hex slotted
Style a - Washer faced
Style b - Double chamfered

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those ligted in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-C-465 - Copper-Aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630, 632M, and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes, and Forgings.
QQ-N-28 1 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, l nd Structural l nd Special Shaped Sections .
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of.

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MILITARY

MIL-T-9047 - Titanium and Titanium Alloy Bars (Rolled or Forged) and Reforging Stock, Aircraft Quality.
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No Go).
MIL-N-24106 - Nickel-Copper Alloy Bars, Rods, and Forgings.
MIL-C-L751 - Coating, Metallic-Ceramic.
MIL-C-83488 - Coating, Aluminum, Ion Vapor Deposited.
MIL-C-87115 - Coating, Immersion Zinc Flake/Chromate Dispersion.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/20 - Screw-Thread Standards for Federal Services Section 20 Inspection Methods for Acceptability of UN, UNR, UNJ, M, and MJ Screw-Threads.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-1312-5 - Fastener Test Methods, Method 5, Stress Durability.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UHR Thread Form).
B1.2 - Gages and Gaging for Unified Inch Screw Threads.
B18.2.1 - Square and Hex Bolts and Screws, Inch Series Including Hex Cap Screws and Lag Screws. (DoD adopted)
B18.2.2 - Square and Hex Nuts. (DoD adopted)
B18.3 - Socket Cap, Shoulder and Set Screws - Inch Series Including Dimensions of Hexagon and Spline Sockets and Keys to Match. (DoD adopted)
B18.5 - Round Read Bolts (Inch Series). (MD adopted)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017 or the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING MD MATERIALS (ASRM)

A 193 - Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service. (DoD adopted)
A 194 - Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service. (DoD adopted)
A 320 - Standard Specification for Alloy Steel Bolting Materials for Low-Temperature Service. (DoD adopted)
A 370 - Standard Methods and Definitions for Mechanical Testing of Steel Products. (DoD adopted)
A 380 - Standard Recommended Practice for Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems. (DoD adopted)
A 574 - Standard Specification for Alloy Steel Socket-Head Cap Screws.
A 751 - Standard Methods, Practices, and Definitions for Chemical Analysis of Steel Products.
B 138 - Standard Specification for Manganese Bronze Rod, Bar, and Shapes. (DoD adopted)
B 139 - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes. (DoD adopted)
B 487 - Standard Method for Measurement of Metal and Oxide Coating Thicknesses by Microscopical Examination of a Cross Section. (DoD adopted)
B 580 - Standard Specification for Anodic Oxide Coatings on Aluminum. (DoD adopted)
B 633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)
B 695 - Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
F 467 - Standard Specification for Nonferrous Nuts for General Use.
F 468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use.
F 593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs. (DoD adopted)
F 594 - Standard Specification for Stainless Steel Nuts. (DoD adopted)
F 606 - Standard Method for Conducting Tests to Determine the Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, and Rivets.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J 121 - Decarburization in Hardened and Tempered Threaded Fasteners, SAE Recommended Practice.
J 122 - Surface Discontinuities on Nuts, SAE Recommended Practice.
J 123 - Surface Discontinuities on Bolts, Screws and Studs, SAE Recommended Practice.
J 429 - Mechanical and Material Requirements for Externally Threaded Fasteners, SAE Standard.
J 995 - Mechanical and Material Requirements for Steel Nuts, SAE Standard.

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.1 Material. Fasteners shall be made of carbon, alloy, corrosion-resistant steel, or nonferrous alloys which conform to table I, as specified (see 6.2.1).

TABLE I. Material grades, chemical and identification requirements.

Material type	Material grade	Fastener type	Chemical requirement Government/industry standards	Identification marking
Carbon and alloy steels	2	Stud, bolt, hex cap screw	SAE J 429	1/
	5	Stud, bolt, hex cap screw	SAE J 429	2/
	8	Stud, bolt, hex cap screw	SAE J 429	2/
	2	Nut	SAE J 995	1/
	5	Nut	SAE J 995	2/
	8	Nut	SAE J 995	2/
5/ Alloy steels	B7	Stud, bolt, hex cap screw	ASTM A 193	B7
	B16	Stud, bolt, hex cap screw	ASTM A 193	B16
	574	Socket head cap screw	ASTM A 574	1/
	4340	Socket head cap screw	ASTM A 574	4340
	L7	Stud, bolt, hex cap screw	ASTM A 320	L7
	L43	Stud, bolt, hex cap screw	ASTM A 320	L43
	2H	Nut	ASTM A 194	2H
	4	Nut	ASTM A 194	4
9/	7	Nut	ASTM A 194	7

See footnotes at end of table.

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TABLE I. Material grades, chemical and identification requirements. - Continued

Material type		Material grade	Fastener type	Chemical requirement Government/industry standards	Identification marking
Corrosion resistant steels	Austenitic 6/	303	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	303 3/ 4/
		303 Se	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	303 Se 3/ 4/
		304	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	304 3/ 4/
		305	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	305 3/ 4/
		316	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	316 3/ 4/
		321	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	321 3/ 4/
		347	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	347 3/ 4/
		384	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	384 3/ 4/
	Martensitic 6/	410	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	410 H 410 HT 3/ 5/
		416, 416 Se	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	416 H, 416 HT, 416 SeH, 416 SeHT 3/ 5/
		431	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	431 H, 431 HT 3/ 5/
	Age hardened	630	Stud, bolt, hex cap screw, socket head cap screw	ASTM F 593	630 3/

See footnotes at end of table.

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TABLE I. Material grades, chemical and identification requirements. - Continued

Material type		Material grade	Fastener type	Chemical requirement Government/industry standards	Identification marking
Corrosion resistant steels	Austenitic 6/	303	Nut	ASTM F 594	303 3/ 4/
		303 Se	Nut	ASTM F 594	303 Se 3/ 4/
		304	Nut	ASTM F 594	304 3/ 4/
		305	Nut	ASTM F 594	305 3/ 4/
		384	Nut	ASTM F 594	384 3/ 4/
		316	Nut	ASTM F 594	316 3/ 4/
		321	Nut	ASTM F 594	321 3/ 4/
		347	Nut	ASTM F 594	347 3/ 4/
	Martensitic 6/	410	Nut	ASTM F 594	410 H, 410 HT 3/ 5/
		416	Nut	ASTM F 594	416 H, 416 HT 3/ 5/
		416 Se	Nut	ASTM F 594	416 SeH, 416 SeHT 3/ 5/
		431	Nut	ASTM F 594	431 H, 431 HT 3/ 5/
	Age hardened	630	Nut	ASTM F 594	630 3/
Nickel alloys	Ni-Cu	400	Stud, bolt, hex cap screw, socket head cap screw	QQ-N-281, class A	NC
	Ni-Cu-Al	500	Stud, bolt, hex cap screw, socket head cap screw	QQ-N-286, class A	"K" 8/
	Ni-Cu 7/	400	Nut	QQ-N-281, class A	NC
		405	Nut	QQ-N-281, class B	NC-R
	Ni-Cu-Al	500	Nut	QQ-N-286, class A	"K"
Copper alloys	Naval brass 6/	462	Stud, bolt, hex cap screw	ASTM F 468	462 3/
		464	Stud, bolt, hex cap screw	ASTM F 468	464 3/
		482	Stud, bolt, hex cap screw, nut	QQ-B-637	482 3/
	Phosphor bronze 6/	510	Stud, bolt, hex cap screw	ASTM F 468	510 3/
		544	Stud, bolt, hex cap screw, nut	ASTM B 139	544 3/

See footnotes at end of table.

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TABLE I. Material grades, chemical and identification requirements. - Continued

Material type		Material grade	Fastener type	Chemical requirement Government/industry standards	Identification marking
Copper alloys	Nickel-aluminum bronze	632	Stud, bolt, hex cap screw, nut	QQ-C-465	632 3/
	Silicon bronze 6/	655	Stud, bolt, hex cap screw	ASTM F 468	655 3/
		661	Stud, bolt, hex cap screw	ASTM F 468	661 3/
	Manganese bronze 6/	670	Stud, bolt, hex cap screw, nut	ASTM B 138	670 3/
		675	Stud, bolt, hex cap screw	ASTM F 468	675 3/
	Naval brass	462	Nut	ASTM F 467	462 3/
		464	Nut	ASTM F 467	464 3/
	Phosphor bronze	510	Nut	ASTM F 467	510 3/
	Silicon bronze	655	Nut	ASTM F 467	655 3/
		661	Nut	ASTM F 467	661 3/
	Manganese bronze	675	Nut	ASTM F 467	675 3/
Titanium alloy		T-7	Stud, bolt, hex cap screw, nut	MIL-T-9047, comp 7	T7
Aluminum alloys		2024	Stud, bolt, hex cap screw	ASTM F 468	2024 3/
		6061	Stud, bolt, hex cap screw	ASTM F 468	6061 3/
		7075	Stud, bolt, hex cap screw	ASTM F 468	7075 3/
Aluminum alloys		2024	Nuts	ASTM F 467	2024 3/
		6061	Nut	ASTM F 467	6061 3/
		7075	Nut	ASTM F 467	7075 3/

See footnotes at top of next page.

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1/ Individual fastener marking to identify the material grade is not required.
2/ Grade 5 shall be marked with three radial lines equally spaced for externally threaded fasteners and three circumferential dashes equally spaced for nuts. Grade 8 shall be marked with six radial lines equally spaced for externally threaded fasteners and six circumferential dashes equally spaced for nuts.
3/ When specified (see 3.9 and 6.2.1) each fastener shall be marked with the material grade.
4/ In addition to the material grade, the marking shall include the heat treatment symbol "'An" when the fastener is machined from annealed stock or when the fastener is annealed after cold forming (see tables II, III and IV).
5/ Suitable nuts for B7 externally threaded fasteners are grades 2H, 4 and 7, and suitable nuts for B16 externally threaded fasteners are grades 4 and 7.
6/ The following groups of material are considered equivalent and interchangeable. When approved by the contracting activity, the contractor may select any alloy within each group:

(a) 303, 303 Se, 304, 305, 384
(b) 32L, 347
(c) 410, 416, 416 Se, 431
(d) 462, 464, 482
(e) 510, 544
(f) 651, 655, 661
(g) 670, 675

7/ The following groups of material are considered equivalent and interchangeable. When approved by the contracting activity, the contractor may select any alloy within each group:

(a) 400, 405

8/ Heat or lot number and the manufacturer's symbol are required (sac 3.9) for fasteners 1/2 Inch in diameter and larger. Only the material symbol shall be required for fasteners less than 1/2 inch in diameter. Numbers used for material identification (see material grade as specified herein) shall not be used for heat or lot number.
9/ Grade 4 and 7 nuts are considered equivalent and interchangeable. When approved by the contracting activity, the contractor may select either grade.

3.1.1 Bar stock. Bar stock for fasteners shall be manufactured using processes specified in the applicable industry or Government standard as shown in table 1.

3.1.2 Special requirements for nickel-copper alloys, grades 500 and 400.

3.1.2.1 Grade 500 nickel alloy externally threaded fasteners. Grade 500 nickel alloy fasteners shall be fabricated by one of the following procedures:

(a) When fasteners are to be machined from bar stock or produced on a bolt maker by cold heading and reduction of the shank, the starting material shall be In the cold drawn and annealed, hot finished and annealed, or hot forged and annealed condition. Threading may be by machining (cut) or rolling. The finished fastener shall be annealed and age hardened.

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(b) When fasteners are produced by hot heading, the starting material may be cold drawn or hot finished and may be in the annealed, annealed and age hardened, or age hardened condition. Threads may be rolled (except on age hardened material) or cut. After forming (and thread rolling), fasteners shall be annealed and age hardened.

(c) Fasteners machined from annealed and age hardened material shall have threads formed by cutting.

3.1.2.2 Grade 400 externally threaded fasteners. Grade 400 externally threaded fasteners shall either be fabricated from hot finished bar stock or cold drawn and stress relieved in accordance with QQ-N-281. When specified (see 6.2.1), for critical systems involving seawater submergence pressure, fasteners shall be cold drawn and stress relieved in accordance with MIL-N-24106.

3.2 Heat treatment. Unless otherwise specified (see 6.2.1), fasteners shall be heat treated in accordance with the applicable document indicated in table I except for martensitic corrosion resistant steel grades 410 and 416 which shall be tempered for 1 hour per inch of diameter but not less than 1 hour.

3.3 Magnetic permeability. For fastener applications where nonmagnetic properties of annealed 300 series corrosion resistant steel are a critical material characteristic, the magnetic permeability test should be specified. When specified (see 6.2.1), the magnetic permeability of annealed corrosion resistant steel fasteners In the austenitic grades (300 series) shall not exceed 2.0. The fasteners may be machined from annealed bar stock or fabricated from cold rolled bar stock and then annealed.

3.4 Chemical properties. Fasteners shall conform to chemical properties specified in the applicable document indicated in table I.

3.5 Mechanical properties. Studs, bolts, and hex cap screws shall conform to the mechanical requirements specified in table II. Socket head cap screws shall conform to the mechanical requirements specified In table 111. Nuts shall conform to the mechanical requirements specified In table IV.

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TABLE II. Mechanical property requirements for studs, bolts, and hex cap screws.

	Grade	Heat treatment or condition 1/	Nominal diameter (inches)	Full-size fasteners			Rockwell hardness	Machined specimens from fasteners or on parent stock			
				Tensile strength ksi	Yield strength ksi min	Proof stress ksi min		Tensile strength ksi min	Yield strength ksi min	Elongation in 4D percent min	Reduction of area percent min
Carbon and alloy steels	2	----	1/4 to 3/4	75-100	----	55	B80 min	75	55	18	35
			Over 3/4	60-100	----	35	B70 min	60	35	18	35
	5	Hardened and tempered	1/4 to 1	120-150	----	85	C23 - 34	120	92	14	35
			Over 1 to 1-1/2	105-150	----	75	B97- C34	105	81	14	35
			Over 1-1/2	90-150	----	55	B90 min	90	58	14	35
	8	Hardened and tempered	1/4 to 2-1/2	150-170	----	120	C32 - 38	150	130	14	35
			Over 2-1/2	140-170	----	105	C30 - 38	140	115	14	35
	B7, L7, L43	Hardened and tempered	1/4 to 2-1/2	125-150	----	95	C25 - 34	125	105	16	50
			Over 2-1/2	115-150	----	85	C22 - 34	115	95	16	50
	B16	Hardened and tempered	1/4 to 2-1/2	125-150	----	95	C25 - 34	125	105	18	50
			Over 2-1/2	115-150	----	85	C22 - 34	115	95	17	45
Corrosion resistant steels	303, 303 Se, 304, 305, 316, 321, 347, 384	Annealed	1/4 and over	75-100	30	----	B95 max	75	30	20	----
		Cold worked	1/4 to 3/4	105-140	65	----	----	105	65	20	----
			Over 3/4	90-140	50	----	----	85	40	25	----
	410, 416, 431, 416 Se	H	1/4 and over	125-150	95	----	C25 - 34	125	95	20	----
		HT	1/4 and over	180-220	135	----	C38 - 47	180	135	12	----

See footnotes at end of table.

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TABLE II. Mechanical property requirements for studs, bolts, and hex cap screws. - Continued

	Grade	Heat treatment or condition 1/	Nominal diameter (inches)	Full-size fasteners			Rockwell hardness	Machined specimens from fasteners or on parent stock			
				Tensile strength ksi	Yield strength ksi min	Proof stress ksi min		Tensile strength ksi min	Yield strength ksi min	Elongation in 4D percent min	Reduction of area percent min
	630	Annealed and age hardened	1/4 and over	135-170	105	----	C28 - 38	135	105	16	----
Aluminum alloys	2024	T4	1/4 and over	62 min	40	----	----	62	40	10	----
	6061	T6	1/4 and over	42 min	35	----	----	42	35	10	----
	7075	T73	1/4 and over	68 min	50	----	----	68	56	10	----
Copper alloys	462, 464, 482	----	1/4 and over	55-90	25	----	----	55	20	25	----
	510, 544	----	1/4 and over	60-90	35	----	----	60	35	15	----
	632	----	1/4 and over	90-120	35	----	----	90	35	18	----
	655, 661	----	1/4 and over	65-100	35	----	----	65	35	15	----
	675, 670	----	1/4 and over	55-90	20	----	----	55	20	20	----
Nickel alloys	400, 405	----	1/4 and over	80-110	40	----	B80 min	80	40	20	----
	2/ 500	Annealed and age hardened	less than 1	130-180	90	----	C24 min	130	90	20	----
			1 and over	130-180	85	----	C24 min	130	85	20	----

See footnotes at end of table.

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TABLE II. Mechanical property requirements for studs, bolts, and hex cap screws. - Continued

	Grade	Heat treatment or condition 1/	Nominal diameter (inches)	Full-size fasteners			Rockwell hardness	Machined specimens from fasteners or on parent stock			
				Tensile strength ksi	Yield strength ksi min	Proof stress ksi min		Tensile strength ksi min	Yield strength ksi min	Elongation in 4D percent min	Reduction of area percent min
Titanium alloy	T7	Annealed	1/4 to 1-1/2	125 min	115 min	----	----	125	115	12	20
			Over 1-1/2 to 2	120 min	110 min	----	----	120	110	12	20
			Over 2	115 min	105 min	----	----	115	105	12	20
		Solution treated and aged	1/4 to 1	145 min	135	----	----	145	135	12	20
			Over 1	135 min	125	----	----	135	125	12	20

1/ See 3.2.
2/ Grade 500 nickel alloy externally threaded fasteners which have been machined from age hardened bar stock shall be tested as full-size fasteners.

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TABLE III. Mechanical property requirements for socket head cap screws.

Grade	Heat treatment or condition	Nominal diameter (inches)	Full-size fasteners			Rockwell hardness	Machined specimens from fasteners or on parent stock			
			Tensile strength ksi min	Yield strength ksi min	Proof stress ksi min		Tensile strength ksi min	Yield strength ksi min	Elongation in 4D percent min	Reduction of area percent min
574, 4340	Hardened and tempered	1/4 to 1/2	180	155	---	C39-45	180	155	8	33
		over 1/2	170	153	---	C37-45	170	153	8	33
303, 303Se, 304, 305, 316, 321, 347, 384	Annealed	1/4 to 5/8	80	30	---	---	80	30	10	30
		over 5/8	70	26	---	---	70	26	20	30
400	---	1/4 and over	80	40	---	B80 min	80	40	20	---
500 1/	Annealed and age hardened	less than 1	130	90	---	C24 min	130	90	20	---
		1 and over	130	85	---	C24 min	130	85	20	---

1/ Grade 500 nickel alloy externally threaded fasteners which have been machined from age hardened bar stock shall be tested as full-size fasteners.

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TABLE IV. Mechanical properties for nuts.

	Grade	Heat treatment or condition[1/]	Nominal size (inch)	Proof stress for hex nut (ksi min)[2/]	Rockwell hardness
Carbon and alloy steels	2	----	1/4 and over	90	B80 min
	2H	Hardened and tempered	1/4 and over	150	C24 - 38
	4, 7	Hardened and tempered	1/4 and over	150	C24 - 38
	5	Hardened and tempered	1/4 thru 1	120	C23 - 32
			Over 1	105	C19 - 32
	8	Hardened and tempered	1/4 and over	150	C24 - 38
Corrosion resistant steels	303, 303 Se, 304, 305, 316, 321, 347, 384	Annealed	1/4 and over	75	B65 - 95
		Cold worked	1/4 and over	90	B95 min
	410, 416, 416 Se, 431	H	1/4 and over	125	C25 - 34
		HT	1/4 and over	180	C38 - 47
	630	Annealed and age hardened	1/4 and over	135	C28 - 38
Aluminum alloys	2024	T4	1/4 and over	55	B70 min
	6061	T6	1/4 and over	40	B40 min
	7075	T73	1/4 and over	52	B60 min
Copper alloys	462, 464, 482	----	1/4 and over	55	----
	510, 544	----	1/4 and over	60	----
	632	----	1/4 and over	90	----
	655, 661	----	1/4 and over	65	----
	670, 675	----	1/4 and over	55	----

See footnotes at end of table.

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TABLE IV. Mechanical properties for nuts. - Continued

Grade		Heat treatment or condition[1/]	Nominal size (inch)	Proof stress for hex nut (ksi min)[2/]	Rockwell hardness
Nickel alloys	400, 405	----	1/4 and over	80	B80 min
	500	Annealed and age hardened	1/4 and over	130	C24 min
Titanium alloy	T7	Annealed	1/4 and over	120	C26 max
		Solution treated and aged	1/4 and over	140	C26 min

1/ See 3.2.
2/ In order to determine minimum proof stress for nuts other than hex nut, use the following:

(a) Proof stress for jam nut (all materials) = 0.60F.
(b) Proof stress for heavy hex nut (ferrous) = 1.10F.
(c) Proof stress for heavy hex nut (nonferrous) = 1.08F.
(d) Proof stress for slotted nut (all materials) = 0.80F.

(Where F = proof stress for hex nut)

3.5.1 Loading stress calculation. Loading stresses for various fastener sizes are contained in most of the applicable ASTM standards listed in table I. Where this information is not available, the values may be calculated by using the following formula:

$$F \text{ (in pounds)} = P \times A_{(s)}$$

Where P = Minimum proof yield or tensile stress, as applicable, of the fastener in pounds per square inch

$$A_{(s)} = 0.7854 \left[D - \frac{0.9743}{n} \right]^2$$

Where $A_{(s)}$ = Stress area of the fastener

D . Nominal diameter or size of fastener
n = Number of threads per inch

The stress areas of externally threaded fasteners and nuts are contained in table V.

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TABLE V. Stress areas of fasteners.

Nominal diameter inches	Coarse thread series (UNC)		Fine thread series (UNF)		8-Thread series (8UN)	
	Threads per inch	Stress area (in^2) A(s)	Threads per inch	Stress area (in^2) A(s)	Threads per inch	Stress area (in^2) A(s)
1/4	20	0.0318	28	0.0364	—	—
5/16	18	.0524	24	.0580	—	—
3/8	16	.0775	24	.0878	—	—
7/16	14	.1063	20	.1187	—	—
1/2	13	.1419	20	.1599	—	—
9/16	12	.182	18	.203	—	—
5/8	11	.226	18	.256	—	—
3/4	10	.334	16	.373	—	—
7/8	9	.462	14	.509	—	—
1	8	.606	12	.663	8	0.606
1-1/8	7	.763	12	.856	8	.790
1-1/4	7	.969	12	1.073	8	1.000
1-3/8	6	1.155	12	1.315	8	1.233
1-1/2	6	1.405	12	1.581	8	1.492
1-5/8	—	—	—	—	8	1.78
1-3/4	5	1.90	—	—	8	2.08
1-7/8	—	—	—	—	8	2.41
2	4-1/2	2.50	—	—	8	2.77
2-1/4	4-1/2	3.25	—	—	8	3.56
2-1/2	4	4.00	—	—	8	4.44
2-3/4	4	4.93	—	—	8	5.43
3	4	5.97	—	—	8	6.51
3-1/4	4	7.10	—	—	8	7.69
3-1/2	4	8.33	—	—	8	8.96
3-3/4	4	9.66	—	—	8	10.34
4	4	11.08	—	—	8	11.81

3.6 Decarburization. Unless otherwise specified (see 6.2.1), the threaded portion of carbon and alloy steel studs, bolts and screws in grades 5, 8, B7, L7, L43, and B16 shall meet the requirements of SAE J 121, class B.

3.7 Protective coating or treatment.

3.7.1 Carbon and alloy steels. When protective coating is specified (see 4.4.7, 6.1.1 and 6.2.1), carbon and alloy steal fasteners shall be coated as follows:

Metallic ceramic coating: MIL-C-81751, type I, class 4.
Ion vapor deposited aluminum coating: ML-C-83488, type II, class 3.
Immersion zinc flake/chromate dispersion coating: MIL-C-87115, class 3.
Electrodeposited zinc coating: ASTM B 633, type II class Fe/Zn 13.
Mechanically deposited zinc coating: ASTM B 695, class 12.

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3.7.1.1 Unless the particular zinc coating method is specified (see 6.2.1), the coating may be applied by the electrodeposited or mechanical method at the option of the manufacturer.

3.7.1.2 Protective coating thickness. The minimum thickness requirements for protective coating shall be as follows:

Type of coating	Coating thickness (rein)
MIL-C-81751, type I, class 4	0.0003 inch 1/
MIL-C-83488, type II, class 3	0.0003 inch
MIL-C-87115, Class 3	0.0003 inch
ASTM B 580, type D	0.0004 inch
ASTM B 633, type II, class 13	0.0005 inch
ASTM B 695, class 12	0,0005 inch

1/ The coating system shall consist of a base coat to MIL-C-81751 type I, class 4, plus an approved phosphate-chromate topcoat. The 0.0003 inch thickness (base coat plus topcoat) is for threaded areas only. Non-threaded areas shall be 0.0015 to 0.0040 inch as specified in the coating specification. Users shall be advised that the 0.0003 inch coating thickness provides marginal corrosion resistance when exposed to a 1000 hour salt spray test.

3.7.1.3 For electrodeposited zinc coating, steel fasteners having a hardness of Rockwell C34 or greater shall require hydrogen embrittlement relief in accordance with ASTM B 633. All other coating types that utilize pickling acids for cleaning or any other process that may introduce hydrogen shall also be subjected to the relief treatment contained in ASTM B 633.

3.7.1.4 Externally threaded coated fasteners. Unless otherwise specified, after coating cLass 2A external threads (see 3.7.1.2) shall meet the dimensional limits for class 3A. Class 3A and class 5 interference fit threads shall not be coated.

3.7.1.5 Internally threaded coated fasteners. Standard internal threads provide no allowance for coating thickness. To determine before coating gauging limits for a uniformly coated thread, increase:

(a) The minimum pitch diameter by four times the maximum coating thickness, if specified, or by six times the minimum of nominal coating thickness, if not specified.
(b) The maximum pitch diameter by four times the minimum or nominal coating thickness.
(c) The minimum minor diameter by two times the maximum coating thickness, if specified, or by three times the mini-mum or nominal coating thickness, if not specified.
(d) The maximum diameter by two times the minimum or nominal coating thickness.

3.7.2 Corrosion-resistant steels. Corrosion-resistant steel fasteners shall be descaled and cleaned In accordance with ASTM A 380.

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3.7.3 Aluminum alloys. When specified (see 4.4.7, 6.1.1, and 6.2.1) aluminum alloy fasteners shall be treated or coated as follows:

Annodized: ASTM B 580, type D
Metallic ceramic coating: MIL-C-81751, type 1, class 4
Ion vapor deposited aluminum: MIL-C-83488, type II, class 3
Immersion zinc flake/chromate dispersion coating: MIL-C-87115, class 3

3.8 Dimensions.

3.8.1 Threads. When specified (see 6.2.1), except for stud end threads of studs with style a or b, screw threads shall be unified form class 2A/2B in accordance with ANSI B1.1. External threads with rounded roots are often identified as UNR (UNRC, UNRF, 8UNR) in accordance with ANSI B1.1 and are acceptable instead of the UN (UNC, UNF, 8UN) series. Thread series shall be coarse for sizes 1/4 through 1 inch and constant 8 thread for sizes over 1 inch, unless otherwise specified. Unless otherwise specified (see 6.2.1), external class 5 interference fit threads shall be class 5HF in accordance with FED-STD-H28. Coated fastener dimensions shall be in accordance with 3.7.1.4 and 3.7.1.5.

3.8.2 Studs.

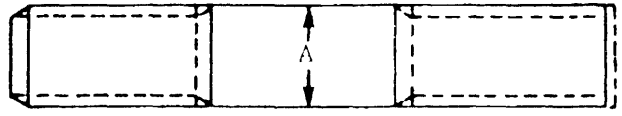
3.8.2.1 Types X, II and III. Body diameter and thread length dimension shall be as shown in tables VI and VII (see 6.2.1). The body diameters of type 1, II and III studs are defined as follows:

(a) Full body. The body diameter (A) is not less than the minimum major diameter of the thread of the nut end, nor more than the nominal (basic) size plus 0.020 inch.
(b) Reduced body. The body diameter (A) is not less than the minimum pitch diameter, nor more than the maximum major diameter of the nut end thread.
(c) Constant strength body.

(1) Cut threads. The body diameter (A) is not less than the stress area diameter nor more than the stress area diameter plus the pitch diameter tolerance for the class fit of the nut end thread.
(2) Rolled threads. The body diameter (A) is the same diameter as the blank before thread rolling. This blank diameter shall fall within the body diameter dimensions shown in table VI.

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TABLE VI. Body diameter - studs, types I, II, and III, styles a, b, c, and d.

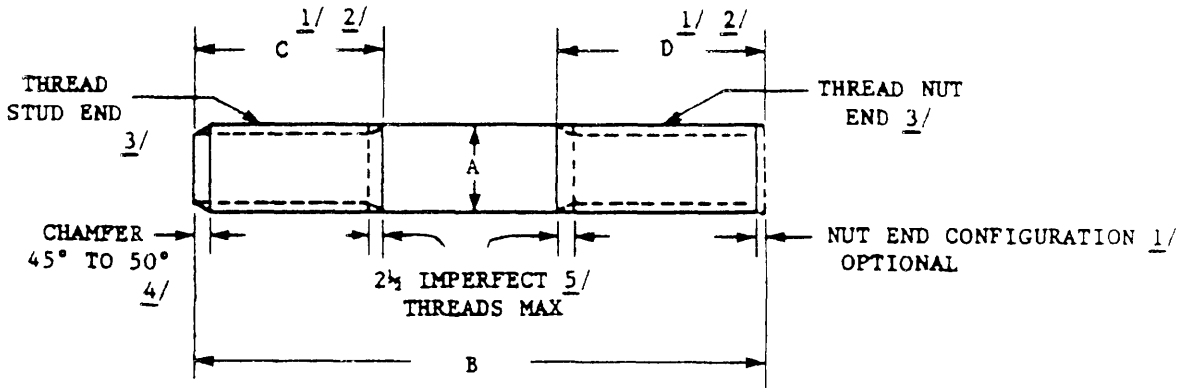


Size	Body diameter (inches)(A)												Nut end thread
	Full body				Reduced body				Constant strength body (type III)				
	Type I Styles a & c		Type I Styles b & d		Type II Styles a & c		Type II Styles b & d		Cut thread 1/ Styles b & d		Rolled thread Styles a, b, c, d		
	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	
1/4	0.270	0.241	0.270	0.240	0.250	0.214	0.248	0.212	0.205	0.201	0.220	0.212	UNC
5/16	.332	.303	.332	.302	.312	.273	.311	.271	.262	.258	.278	.271	UNC
3/8	.395	.365	.395	.364	.375	.331	.373	.328	.318	.314	.336	.328	UNC
7/16	.457	.427	.457	.425	.437	.387	.436	.385	.373	.368	.393	.385	UNC
1/2	.520	.489	.520	.487	.500	.446	.498	.443	.430	.425	.452	.443	UNC
9/16	.582	.551	.582	.549	.562	.504	.560	.501	.487	.481	.510	.501	UNC
5/8	.645	.612	.645	.611	.625	.561	.623	.558	.542	.536	.568	.558	UNC
3/4	.770	.737	.770	.735	.750	.680	.748	.677	.658	.652	.687	.677	UNC
7/8	.895	.861	.895	.859	.875	.798	.873	.794	.773	.767	.804	.794	UNC
1	1.020	.985	1.020	.983	1.000	.913	.998	.910	.885	.878	.920	.910	UNC
1-1/8	1.145	1.108	1.145	1.106	1.125	1.026	1.122	1.022	.993	.986	1.034	1.022	UNC
1-1/8	1.145	1.110	1.145	1.108	1.125	1.039	1.122	1.035	1.010	1.003	1.046	1.034	8UN
1-1/4	1.270	1.233	1.270	1.231	1.250	1.151	1.247	1.147	1.118	1.111	1.159	1.147	UNC
1-1/4	1.270	1.235	1.270	1.233	1.250	1.164	1.247	1.160	1.135	1.129	1.171	1.159	8UN
1-3/8	1.295	1.356	1.295	1.354	1.375	1.260	1.372	1.256	1.221	1.213	1.269	1.256	UNC
1-3/8	1.295	1.360	1.295	1.358	1.375	1.288	1.372	1.284	1.260	1.253	1.296	1.284	8UN
1-1/2	1.520	1.481	1.520	1.479	1.500	1.385	1.497	1.381	1.346	1.338	1.394	1.381	UNC
1-1/2	1.520	1.485	1.520	1.483	1.500	1.413	1.497	1.409	1.386	1.378	1.421	1.409	8UN

1/ Maximum values are for class 2A fit only. 3A fit should use 3A pitch diameter tolerance limits as found in FED-STD-H28/20.

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TABLE VII. Thread length dimensions for studs, types I, II, and III, styles a, b, c, and d.



Nominal size			Minimum thread length (inches)		
	T.p.i.		Nut end (D)		
Inch	Coarse thread series	Fine thread series	Up to 6 inches long (B)	Over 6 inches long (B)	Stud end (C)
1/4	20	28	0.750	1.000	0.375
5/16	18	24	.875	1.125	.460
3/8	16	24	1.000	1.250	.562
7/16	14	20	1.125	1.375	.656
1/2	13	20	1.250	1.500	.750
9/16	12	18	1.375	1.625	.844
5/8	11	18	1.500	1.750	.937
3/4	10	16	1.750	2.000	1.125
7/8	9	14	2.000	2.250	1.312
1	8	12	2.250	2.500	1.500
1-1/8	7	12	2.500	2.750	1.687
1-1/4	7	12	2.750	3.000	1.875
1-3/8	6	12	3.000	3.250	2.062
1-1/2	6	12	3.250	3.500	2.500

1/ When studs are too short to be threaded as specified, the threading shall be of equal length from each end to the center and the ends shall be rounded or chamfered.

2/ The maximum thread length shall be specified minimum length plus 3/16 inch or 2-1/2 threads, whichever is greater.

3/ Thread taper variations shall fall within the applicable pitch diameter tolerances.

4/ The stud end of studs shall be chamfered 45 to 50 degrees from a point 1/64 to 1/32 inch below the root diameter of the thread. The shape of the nut end is optional except as specified in 1/, but shall fall within the same dimensional envelope as the stud end.

5/ Imperfect or incomplete threads at the junction of the two thread classes shall not exceed 2-1/2 threads and shall occur in the nut end.

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3.8.2.2 Type IV. Diameter and length of continuous thread studs, type IV, shall be as specified (see 6.2.1).

3.8.2.3 Straightness. When rolled on a flat surface, the clearance as measured by a feeler gauge shall not exceed the value BT/5D, where B is the nominal length, D the nominal diameter, and T is the pitch diameter tolerance of the nut end thread.

3.8.2.4 Length tolerance. Length tolerance for studs shall be as follows:

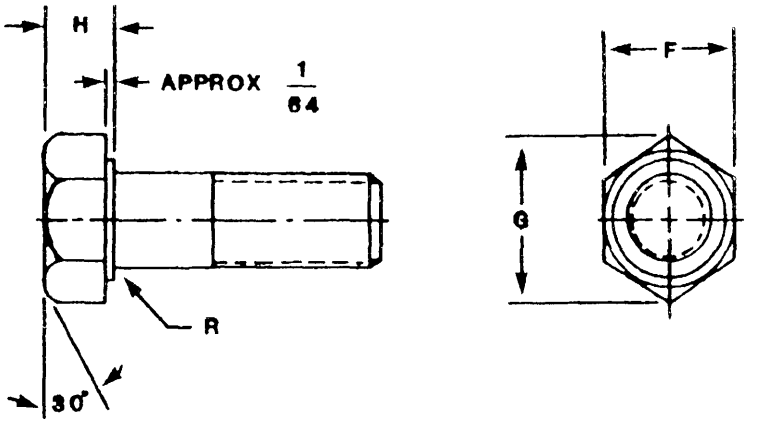
Fastener length (B)	Tolerance for nominal stud diameter			
	1/4 to 3/8	7/16 to 1/2	5/8 to 1-1/4	Over 1-1/4
Up to 6 inches	±1/32	±1/16	±1/8	±1/4
Over 6 inches	±1/16	±3/32	±3/16	±1/4

3.8.3 Bolts. Dimensions of type I, 11, and 111 bolts shall conform to ANSI B18.2.1. Dimensions of types IV and V bolts shall conform to ANSI B18.5. Unless otherwise specified (see 6.2.1), the bearing surface shall be either washer faced or chamfered.

3.8.4 Hex cap screws. Dimensions of type I hex cap screws (finished hex bolts) in nominal diameters of 1/4 through 3 inches shall conform to ANSI B18.2.1. Dimensions of type I hex cap screws in nominal diameter of 3-1/4 through 4 inches shall be as shown in table VIII.

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TABLE VIII. Type I hex cap screw (sizes 3-1/4 to 4 inches).



Nominal size or basic major diameter of thread		Body diameter 1/	Width across flats F			Width across corners G		Height H			Radius of fillet R	
		Max	Max (basic)		Min	Max	Min	Nominal	Max	Min	Min	Max
3-1/4	3.2500	3.335	4-7/8	4.8750	4.712	5.629	5.372	2	2.064	1.936	0.125	0.188
3-1/2	3.5000	3.589	5-1/4	5.2500	5.075	6.062	5.786	2-1/8	2.193	2.057	0.125	0.188
3-3/4	3.7500	3.858	5-5/8	5.6250	5.437	6.495	6.198	2-5/16	2.385	2.241	0.125	0.188
4	4.0000	4.111	6	6.0000	5.800	6.928	6.612	2-1/2	2.576	2.424	0.125	0.188

1/ There may be a reasonable swell or fin under the head or die seam on the body not to exceed the nominal body diameter by the following:

0.183 for sizes over 3 inches

NOTES:

1. Dimensions given in inches.
2. Notes pertaining to hex cap screws contained in ANSI B18.2.1 shall apply.

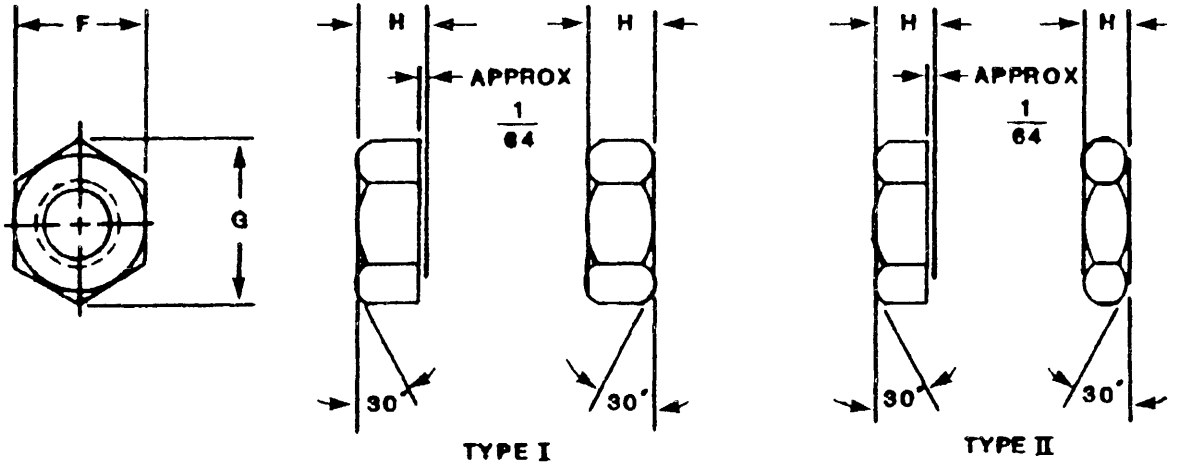
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3.8.5 Socket head cap screws. Dimensions of hexagon socket head cap screws shall conform to ANSI B18.30

3.8.6 Nuts. Dimensions of type I, II, III, IV and V nuts shall conform to ANSI B18.2.2. Unless otherwise specified (see 6.2.1), the bearing surface shall be either washer faced or chamfered. Dimensions of type I hex nuts (large diameter) and type II hex jam nuts (large diameter) in nominal diameters of 1-3/4 through 3 inches shall be as shown in table IX. Dimensions of type V hex slotted nuts (large diameter) in nominal diameters of 1-3/4 through 3 inches shall be as shown in table X.

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TABLE IX. Type I hex nut and type II hex jam nut (size 1-3/4 to 3 inches).



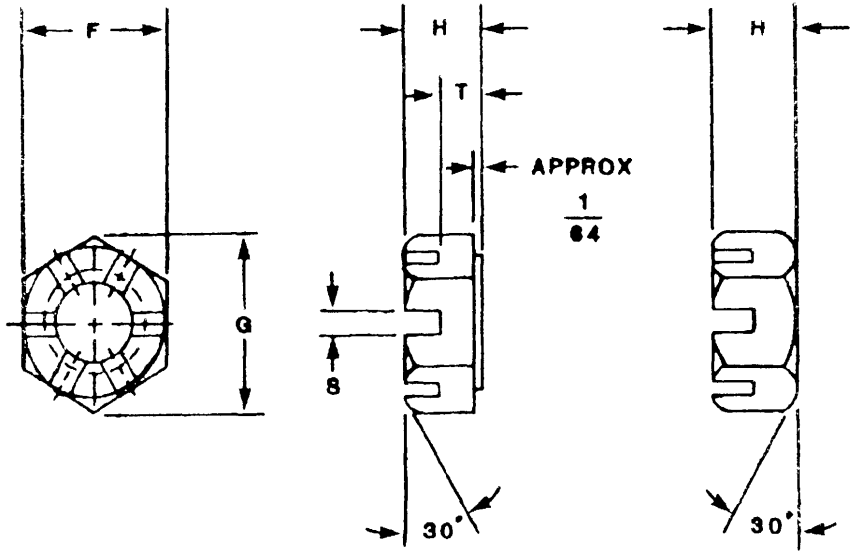
Nominal size or basic major diameter of thread		Width across flats F			Width across corners G		Thickness nuts H			Thickness jam nuts H		
		Max (basic)		Min	Max	Min	Nominal	Max	Min	Nominal	Max	Min
1-3/4	1.7500	2-5/8	2.6250	2.538	3.031	2.893	1-1/2	1.540	1.460	31/32	1.009	0.929
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	1-3/32	1.138	1.050
2-1/4	2.2500	3-3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	1-13/64	1.251	1.155
2-1/2	2.5000	3-3/4	3.7500	3.625	4.330	4.133	2-9/64	2.193	2.089	1-29/64	1.505	1.401
2-3/4	2.7500	4-1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	1-37/64	1.634	1.522
3	3.0000	4-1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	1-45/64	1.763	1.643

NOTES:

1. Dimensions given in inches.
2. Notes pertaining to hex nuts and hex jam nuts contained in ANSI B18.2.2 shall apply.

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TABLE X. Type V hex slotted nut (large diameter).



Nominal size or basic major diameter of thread		Width across flats F			Width across corners G			Thickness nuts H		Slot T			
		Max (basic)		Min	Max	Min	Nominal	Max	Min	Width S		Unslotted thickness	
										Max	Min	Max	Min
1-3/4	1.7500	2-5/8	2.6250	2.538	3.031	2.893	1-1/2	1.540	1.460	0.52	0.43	1.24	1.20
2	2.0000	3	3.0000	2.900	3.464	3.306	1-23/32	1.763	1.675	0.52	0.43	1.43	1.38
2-1/4	2.2500	3-3/8	3.3750	3.262	3.897	3.719	1-59/64	1.970	1.874	0.52	0.43	1.67	1.62
2-1/2	2.5000	3-3/4	3.7500	3.625	4.330	4.133	2-9/64	2.193	2.089	0.64	0.55	1.79	1.74
2-3/4	2.7500	4-1/8	4.1250	3.988	4.763	4.546	2-23/64	2.415	2.303	0.64	0.55	2.05	1.99?
3	3.0000	4-1/2	4.5000	4.350	5.196	4.959	2-37/64	2.638	2.518	0.71	0.62	2.23	2.19?

NOTES:

1. Dimensions given in inches.
2. Notes pertaining to hex slotted nuts contained in ANSI B18.2.2 shall apply.

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3.9 Identification marking. Unless otherwise specified (see 6.2.1), identification symbols/markings shall be as shown in table I. When heat-to-heat traceability (lot definition "a") is desired and for grade 500 nickel alloy fasteners, the manufacturer's symbol, material synbol, and lot number shall be marked on fasteners 1/2 inch in diameter or larger. For fasteners less than 1/2 inch in diameter, only the material symbol shall be marked. Marking shall be permanent and may be raised or depressed. Stamping shall be made with a low stress stamp. Marking shall be applied to the head of bolts and hex cap screws, the threaded end of socket head cap screws, the nut end of type I, 11, and III studs , either end of type IV studs, and the top face of nuts.

3.10 Workmanship.

3.10.1 Fastener quality. Fasteners shall be uniform in quality and condition, free from rust, scale, and burrs, and without broken, burred, or damaged threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all Inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the (government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and cervices conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling in quality conformance does nut authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Insection lot. A lot is defined as either (a) or (b) below (see 6.2.1:

(a) A lot of fasteners consists of one heat of material, of one type and style, of the same nominal size, and fabricated and heat treated in the same batch or by a continuous process under the same conditions as to time and temperature.
(b) A lot of fasteners consists of one grade of material, of one type and style, of the same nominal size, and produced under the same conditions.

4.2.1 For grade 500 and grade T7 materials, lot definition (a) shall apply. For other materials, either (a) or (b) may be specified (see 6.2.1), depending on whether heat-to-heat traceability is desired.

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4.3 Sampling and acceptable quality level (AQL). Unless otherwise specified (see 6.2.1), sampling and AQL shall be as shown in tables XI and XII.

TABLE XI. Sampling and AQL.

Type of test or inspection	Number of samples per lot or method
Chemical analysis Decarburization (see 3.6)	One for each heat or lot as applicable
Dimensions and marking	MIL-STD-105, general inspection level I. AQL as shown in table XII.
Mechanical test 1/ Stress durability testing	2/
Nondestructive testing	MIL-STD-105, special inspection level S-4, AQL = 0.65 (see 4.4.5)
Nondestructive testing Permeability measurement (see 3.3)	MIL-STD-105, general inspection level I, AQL = 2.5
Coating thickness	2/
Dimensions after coating: Pitch and functional diameter	MIL-STD-105, general inspection level I. AQL as shown in table XII.

1/ Mechanical test includes all required tests specified in 4.4.4. All of the required tests shall be performed on each sample. For fasteners which attain their properties through heat treatment, the samples shall be obtained from each furnace load or each continuous process not exceeding 8 hours. For fasteners which are fabricated from hardened material, or which require no heat treatment, the samples shall be obtained from each heat or lot of material (see 4.4.4.6).

2/ Sampling shall be as follows:

Lot size (pc)	Sample size (pc)	Acceptable defects
50 and less	2	0
51 to 500	3	0
501 to 35,000	5	0
Over 35,000	8	0

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TABLE XII. Examination requirements.

Description	Applicable fastener	AQL
Major diameter	Bolts, hex cap screws, studs, socket head cap screw	2.5
Minor diameter	Nuts	2.5
Pitch diameter	Bolts, hex cap screws, studs, nuts, socket head cap screws	2.5
Functional diameter (functional size)	Bolts, hex cap screws, studs, nuts, socket head cap screws	2.5
Identification symbols (if required)	Bolts, hex cap screws, studs, nuts, socket head cap screws	4.0
Thickness	Nuts	4.0
Head height	Bolts, hex cap screws, socket head cap screws	4.0
Head diameter	Socket head cap screws	4.0
Hex socket size	Socket head cap screws	4.0
Key engagement	Socket head cap screws	4.0
Distance across flats	Bolts, hex cap screws, nuts	4.0
Width across corners	Bolts, hex cap screws, nuts	4.0
Total runout (concentricity)	Bolts, hex cap screws, studs, socket head cap screws	4.0
Body diameter	Bolts, hex cap screws, studs, socket head cap screws	4.0
Overall length	Bolts, hex cap screws, studs, socket head cap screws	4.0
Thread length	Bolts, hex cap screws, studs, socket head cap screws	4.0
Fillet radius	Bolts, hex cap screws, socket head cap screws	4.0
Diameter of washer face	Bolts, hex cap screws, nuts, socket head cap screws	4.0
Straightness	Bolts, hex cap screws, studs, socket head cap screws	4.0
Quality (see 3.10.1)	Bolts, hex cap screws, studs, nuts, socket head cap screws	4.0

4.4 **Methods of examination and test.**

4.4.1 Chemical analysis. Chemical analysis shall be performed as specified in the applicable material specification as shown on table I and shall meet the requirements therein. Standard tolerance permitted for product analysis by the applicable specification applies. Certified heat analysis (see 4.5) furnished by the material manufacturer is acceptable instead of product analysis.

4.4.2 Decarburization. Decarburization examination shall be made in accordance with SAE J 121, class B.

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4.4.3 Dimension requirements. Table XII specifies the parameters required for dimensional inspection. Screw thread gauging practices shall be in accordance with FED-STD-H28/20, system 22. Screw thread gauges shall be in accordance with ANSI B1.2. For externally threaded coated fasteners, threads accepted to class 2A limits before coating shall be inspected to class 3A limits after coating.

4.4.4 Mechanical tests. Mechanical tests shall be performed in accordance with ASTM F 606. Tensile tests may be performed on full size fasteners, turned-down shank, or tensile specimens prepared from the parent bar stock. Tensile specimens prepared from fastener or parent bar stock larger than 1-1/2 inches In diameter shall be machined off center In accordance with ASTM F 606. Testing of full size externally threaded fasteners shall include the proof stress or the yield strength and the tensile strength test. Testing of turned down specimens or tensile specimens shall include tensile, yield, elongation and reduction of area as specified in table II. The speed of testing, determined with a free running cross head, shall not exceed 1/8 inch per minute for proof load or yield strength test, and 1 inch per minute for axial tensile or wedge tensile test. If the failure of a test sample is due to improper testing techniques, the test results shall be discarded and a new test substituted. Mechanical properties shall be as shown in tables II, III, and IV. Testing for mechanical properties shall be in accordance with table XIII and 4.4.4.1 through 4.4.4.6.

4.4.4.1 Wedge tensile. Headed fasteners with heads formed by upsetting shall be given a wedge test. For purpose of determining the wedge angle to be used , as specified in ASTM F 606, the bolt or hex cap screw is considered threaded-to-head when the unthreaded portion of the shank, excluding the fillet radius , is 1/4 Inch or less.

4.4.4.2 Proof load for externally threaded fasteners. The proof load test is applicable for carbon and alloy steel externally threaded fasteners (see table II). The proof load test shall be performed in accordance with ASTM F 606, prior to the axial or wedge tensile test.

4.4.4.3 Yield strength. The yield strength test is required for all externally threaded fasteners, except those which have been proof load tested (see table 11).

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TABLE XIII. Mechanical property test requirements.[1/]

Products	Test on full size fasteners				Test on machined[3/] specimen from fastener or parent bar stock
	Hardness[2/]	Proof stress or yield strength	Axial tensile	Wedge tensile	
Bolts and screws	X	Option A	---	X[4/]	Option B
Studs	X	Option A	Option A	---	Option B
Nuts Proof load up to 150,000 pounds	X	X	---	---	---
Proof load over 150,000 pounds	X	5/	---	---	---

1/ The check marks (X) denote mandatory tests. The dash marks (-) denote tests not required. Where options are indicated, either all of the option A tests or the option B tests shall be performed at the discretion of the manufacturer. The results of full size test shall take precedence over machined specimen test results for acceptance purposes where arbitration is necessary.
2/ Hardness test is required only where it is specified in tables II and 111.
3/ See 4.4.4.6.
4/ All headed fasteners with heads formed by upsetting shall be given a wedge tensile test.
5/ Tests shall be performed to the required proof stress. here equipment limits the loading, the test shall be performed to the capacity of the equipment which must be at least 150,000 pounds.

4.4.4.4 Proof load for nuts. Proof load testing is required for all nuts , except those which require a test load greater than 150,000 pounds. Testing shall be conducted in accordance with 4.4.4, using a hardened mandrel or bolt and a load holding time of 10 seconds at the specified proof load. The proof bad shall be calculated by multiplying the appropriate proof stress from table IV by the corresponding stress area from table IV. The nut shall resist the proof load without stripping or rupturing and shall be removable from the test bolt or mandrel by hand. Use of a wrench to start the nut is permitted, provided the nut is turned no more than 180 degrees before removal by hand.

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4.4.4.5 Hardness. Fasteners shall be tested for hardness, in accordance with ASTM A 606 to meet the hardness requirements of table II or III, as appropriate, Hardness conversion in accordance with ASTM A 370 is permitted.

4.4.4.6 Alternate mechanical testing. Certified mechanical test reports (see 4.5) furnished by the bar stock supplier which demonstrates Compliance with the mechanical property requirement of tables 11 and 111 may be substituted for mechanical testing of fasteners provided that:

(a) Bolts. screws, studs or nuts were fabricated solely by machining without further heat-treatment.
(b) Studs were fabricated solely by thread rolling without further heat treatment, except for grade 500 nickel alloy studs manufactured from age hardened bar stock which shall be formed by cutting.
(c) Bolts and screws fabricated by thread rolling and head upsetting, and without further heat treatment, shall require only the wedge tensile test.
(d) Lot definition "'a" is specified and marking or Identification is maintained to ensure traceability to the raw material specification.

4.4.5 Nondestructive testing. Fasteners shall be nondestructively inspected by wet magnetic particle or Liquid penetrant tests in accordance with MIL-STD-271. When protective coating is specified, the nondestructive tests shall be performed before the coating process.

4.4.5.1 Unacceptable defects. Defect definitions, pictorial descriptions, and defect acceptability limits are contained in SAE J 122 for nuts, and SAE J 123 for bolts, screws and studs.

4.4.5.2 Determination of defect depth. Inspected fasteners with defects which require depth determination may be accomplished by macroscopic examination at 25x magnification minimum, micrometer measurement before and after defect removal, or microscopic examination of specimens prepared by sectioning through the defects. Ten percent of the fasteners with seams shall be tested for depth determination. When the sample size is less than one, one sample shall be tested. If none of the samples contains seams that exceed the permissible limits, then the entire lot is considered acceptable.

4.4.6 Magnetic permeability. Determination of permeability shall be conducted by means of permeability indicators in accordance with MIL-I-17214 to meet the requirements of 3.3.

4.4.7 Protective coating thickness. Protective coating thicknesses shall be determined by microscopic examination In accordance with ASTM B 487. A minimum of four measurements shall be made on each sample, and shall Include two readings at the thread pitch diameter locations and two readings at non-threaded sections. Failure of the coating thickness to meet the minimum coating thickness requirements (see 3.7.1.2) shall be cause for lot rejection.

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4.4.8 Stress durability. If coated, the following grades of steel fasteners shall be subjected to a hydrogen embrittlement stress durability test in accordance with MIL-STD-1312, test no. 5: grade 8, 400 series corrosion resistant steels with HT heat treatment, grade 630 corrosion resistant steel, and socket head cap screw grades 574 and 4340. The fasteners shall be held under load for 48 hours.

4.4.9 Marking. Fasteners shall be visually inspected for material identification, when required (see table I, and 6.2.1). Markings which are incorrect, missing or illegible shall constitute a defect for lot rejection purposes.

4.4.10 Impact. Grades L7 and L43 alloy steel fasteners shall impact tested in accordance with ASTM A 320.

4.5 Test reports. When specified in the contract or order, a certified test report for each lot of fasteners shall be prepared (see 6.2.2).

4.6 Inspection packaging. Sample packages and packs and the inspection of the preservation-packaging , packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of the applicability of the packaging requirements of referenced documents listed in section 2, see 6.5).

5.1 Packaging, packing and marking. Fasteners shall be preserved-packaged level A or C packed level A, B, or C as specified (see 6.2.1), and marked in accordance with PPP-H-1581, except that externally threaded fasteners over 5/6 inch nominal diameter shall be furnished with thread protectors or other suitable cushioning.

5.2 Cushioning, dunnage and wrapping materials.

5.2.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill material for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for material destined for shipboard installation and stowage.

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5.2.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing application such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

" CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.2.3 Cushioning. Cushioning, filler, dunnage and wrapping materials selected, whenever available shall exhibit resistance to fire.

6. NOTES

6.1 Intended use. This specification is intended to cover fasteners for critical applications where a high degree of reliability is required. Commercial fastener standards should be specified for general purpose fasteners.

6.1.1 High strength fasteners.

6.1.1.1 Coating of grade 8 steel fasteners and socket head cap screws (grade 574 and 4340) for corrosion protection increases their susceptability to embrittlemnt. The uncoated fastener grades are subject to accelerated corrosion in seawater environments. Therefore, grade 8 steel fasteners and alloy steel socket head capscrews should not be coated due to their susceptability to embrittlement and are not recommended for use in seawater environments. The above requirements on coating and use in seawater environments do not apply to fasteners used as replacements for FF-S-86 fasteners.

6.1.1.2 Grades 410, 416, and 431 corrosion resistant steels should not be used for seawater environment shipboard applications. These fastener grades are susceptible to both stress corrosion cracking and hydrogen embrittlement due to the hydrogen released in the corrosion reaction when galvanically coupled with steel.

6.1.2 Steel fasteners. The recommended use temperatures for steel fastener grades are contained in table XIV.

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TABLE XIV. Recommended use temperatures of steel fasteners.1/

Steel grades	Use temperature
Carbon steel grades SAE 2 and 5	General use to 650 degrees Fahrenheit (°F)
Grade B7 externally threaded used with grade 2H, 4, or 7 nuts	For use to 775°F fasteners
Grade B16 externally threaded fasteners used with grade 4 or 7 nut	For use to 1000°F
Grade L7 and L43	For low temperature use from minus 60°F to 775°F

1/ The temperatures apply to the fastener materials only and are subject to coating material limitations of table XVI.

TABLE XV. Materials for fasterners exposed to salt water spray.

Metals being joined	Fastener material to be used1/ (in descending priority)	
	Less than 5/8 inch diameter	5/8 inch diameter or greater
Steel and steel CRES and steel	CRES 316 alloy2/ Silicon bronze grades 655, 661 Nickel-copper-aluminum, grade 500	Steel with metallic-ceramic coating CRES 316 alloy2/ Silicon bronze grades 655, 661 Nickel-copper-aluminum, grade 500
CRES and CRES	CRES 316 alloy 2/	CRES 316 alloy2/
Steel and aluminum alloys	CRES 316 alloy2/ Silicon bronze grades 655, 661	Steel with metallic-ceramic coating CRES 316 alloy 2/ Silicon bronze grades 655, 661

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TABLE XV. Materials for fasterners exposed to salt water spray. - Continued

Metals being joined	Fastener material to be used 1/ (in descending priority)	
	Less than 5/8 inch diameter	5/8 inch diameter or greater
Aluminum and aluminum alloys	Aluminum 6061 alloy	Aluminum 6061 alloy Steel with metallic-ceramic coating
Copper and copper alloys	Silicon bronze grades 655, 661 Nickel-copper-aluminum, grade 500	Silicon bronze grades 655, 661 Nickel-copper-aluminum, grade 500

1/ These fastener materials are intended for design selection and should not be substituted In existing applications without load evaluations.
2/ Grade 316 CRES is susceptible to crevice and pitting corrosion when submerged in stagnant seawater and should not be used under these conditions.

6.1.3 Bolting subject to seawater corrosion.

6.1.3.1 Exposure to salt water spray. The recommended materials for fasteners exposed to salt water spray are contained in table XV.

6.1.3.2 Connection submerged in bilge or seawater. Grades 400 and 500 nickel alloy fasteners should be used for connections of material submerged in bilge or seawater and in bilge regions.

6.1.4 Corrosion resistant steal fasteners. Corrosion resistant steel bolts and nuts (300 series) should be used for mounting radio and radar antennas to their foundations and where corrosion resistance and low magnetic properties are desired.

6.1.5 Protective coatings. The characteristics of the various coatings are contained in the applicable coating specifications. Suitable applications are summarized in table XVI.

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TABLE XVI. Coatings for various applications.

Coating type	Base material	Upper temperature 1/ limit	Required salt spray test (min)	Environment
Metallic-ceramic MIL-C-81751, type I, class 4	Carbon and alloy steels 2/ 5/8 inch and larger	1000°F	1000 hours	High temperature and severe corrosion
IVD aluminum MIL-C-83488, type II, class 3	Carbon and alloy steels 2/. No size limitation	925°F	336 hours	Moderate temperature and corrosion
Immersion zinc/ flake MIL-C-87115, class 3	Carbon and alloy steels 2/. No size limitation	650°F	336 hours	Moderate temperature and corrosion
Zinc coating ASTM B 695, class 12 or ASTM B 633, type II, class Fe/Zn 13	Carbon and alloy steels 2/. No size limitation	300°F	96 hours	Ambient temperature and mild corrosion

1/ Maximum service temperature applies to coating alone and not to the base metal (see table XIV).
2/ Grade 8 alloy steel shall not be coated as all coatings are anodic to steel and may produce hydrogen embrittlement in high strength steels.

6.1.6 Identification lot. Users are advised that for level I applications (as defined in NAVSEA 0948-LP-045-7010, appendices A and B) lot definition "a'" shall be used when ordering. Lots procured to definition "b'* may be inhomogeneous with regard to material chemistry and processing except for heat treatment. Therefore, the sampling plan defined in table XII may provide significantly less assurance than lot definition "a".

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, material designation, and heat treatment or condition of fasteners (see 1.2, 3.1, 3.2, and table I).
(c) MIL-N-24106, if specified for grade 400 nickel alloy (see 3.1.2.2).
(d) Permeability test, if required for annealed corrosion resistant (300 series) steel (see 3.3).
(e) Decarburization if other than specified (see 3.6).
(f) Type of coating, if required (see 3.7).

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(g) Size, length (for studs, bolts and screws) thread series, and class of fit (see 3.8.1, 3.8.2.1, 3.8.2.2, 3.8.3, 3.8.4, 3.8.5 and 3.8.6).
(h) Bearing surface (see 3.8.3 and 3.8.6).
(i) Identification, if other than specified (see table I and 3.9).
(j) Whether lot (a) or (b) is required or whether special sampling is required (see 4.2, 4.2.1, and 4.3).
(k) Level of preservation-packaging and packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirements title	Applicable DID no.	Option
4.5	Reports, test	DI-T-2072	---

(Data item descriptions related to this specification, and identifed in section 6 will be approved and listed as such in DoD 5010. 12-L., Vol. 1, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Supersession data. This specification supersedes MIL-B-857A(SHIPS) dated 1 June 1956 with its Amendment 5 dated 28 February 1969, and MS17980(SHIPS) dated 14 May 1963. A cross reference of type designations appears in table XVI.

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TABLE XVI. Type comparability.

Type	Description	MIL-S-1222H	MIL-S-001222G	MIL-S-1222F	MIL-B-857A
Studs	Full body	Type I	Type I	---	Type I
	Reduced body	Type II	Type II	---	Type II
	Constant strength body	Type III	Type III	---	---
	Continuous thread	Type IV	Type IV	Type I (bolt-stud)	---
Bolts	Hex	Type I	Type I	---	Type I
	Heavy hex	Type II	Type II	---	---
	Heavy hex structural	Type III	Type III	---	---
	Round head	Type IV	---	---	Type IV
	Round head square neck	Type V	---	---	Type V
Screws	Hex cap	Type I	Type I	---	Type II, III
	Hex socket head cap	Type II	---	---	---
Nuts	Hex	Type I	Type I	Type II (finished)	Type I, II, III
	Hex jam	Type II	Type II	---	Type IV, V
	Heavy hex	Type III	Type III	Type II (semi-finished)	---
	Heavy hex jam	Type IV	Type IV	---	---
	Hex slotted	Type V	---	---	Type VI, VII

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6.4 Cross-reference of classifications. The fastener types covered by this specification and the corresponding types covered by superseded specifications are listed in table XVI. Except as noted in 6.4.1 and 6.4.2, the fasteners covered by MIL-S-1222H are interchangeable with those covered by the superseded specifications shown in table XVI.

6.4.1 Hex cap screws. The type I finished hex cap screw of MIL-S-1222H is interchangeable with the type III regular finished hexagon head bolt of MIL-B-857A. The type I finished hex cap screw of MIL-S-1222H shall be substituted for the type 11 regular semi-finished hexagon head bolt of MIL-B-857A for nominal sizes up to and including 3 inches. The type I finished hex cap screw of MIL-S-1222H has closer tolerances than the type II regular semi-finished hexagon head bolt of MIL-B-857A for the following dimensions: body diameter, width across flats, width across corners, and radius of fillet. The type II semifinished hex head bolt of MIL-S-1222H is interchangeable with the type II regular semi-finished hexagon head bolt of MIL-B-857A for nominal sizes over 3 inches. Material comparability of MIL-S-1222H and the superseded specifications is contained in table XVII.

TABLE XVII. Material comparability.

	MIL-S-1222H	MIL-S-001222G	MIL-S-1222F	MIL-B-857A
Carbon and alloy steels	Grade	1/ Grade 1 (use grade 2)	Grade	Grade 1
	2	2	—	2
	5	5	--	5
	8	8	—	8
	B7	B7	B7	--
	--	--	B14 (use grade B16)	—
	B16	B16	B16	—
	L7	--	--	--
	L43	—	--	—
	2H	2H	H	—
	4	4	4	—
	1/ 7	--	—	--
Corrosion resistant steel	303	303	—	303
	303 Se	--	--	—
	304	304	—	—
	305	305	—	305
	316	316	—	316
	321	321	—	—
	347	347	—	--
	384	384	--	—
	410	410	—	—
	416	416	--	--
	416 Se	—	—	—
	431	—	—	—
	630	—	—	—

See footnote at end of table.

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TABLE XVII. Material comparability. - Continued

	MIL-S-1222H	MIL-S-001222G	MIL-S-1222F	MIL-B-857A
	Grade	Grade	Grade	Grade
Aluminum alloys	2024	2024		—
	6061	6061		--
	7075	—		--
Copper alloys	462	462		Naval brass
	464	464		Naval brass
	482	482		Naval brass
	510	510		Phosphor bronze
	544	544		Aluminum bronze
	632	632		Ni-Al bronze
	--	651 (use grade 655 or 661)		Silicon bronze
	655	655		Silicon bronze
	661	661		Silicon bronze
	675	675		Manganese bronze
Nickel alloys	400	400		Ni-Cu, class A
	405	405		Ni-Cu, class B
	500	500		Ni-Cu-A2
Titanium alloy	T7	T7		--

1/ May be used interchangeably with grade 4 steel.

6.4.2 Socket head cap screws. The type II socket head cap screw of MIL-S-1222H has no type interchangeability with previous MIL-S-1222 revisions or MIL-B-857A.

6.4.3 Nuts. The type I hex nut of MIL-S-l222H is interchangeable with the type III finished hexagon nut of MIL-B-857A. The type I hex nut of MIL-S-1222H shall be substituted for the type I regular hexagon nut and the type II regular semi-finished hexagon nut of MIL-B-857A. The type I hex nut of MIL-S-1222H has closer tolerances than either the types I or II nuts of MIL-B-857A. The type 11 hex jam nut of MIL-S-1222H is interchangeable with the type V finished hexagon jam nut of MIL-B-857A. The type 11 hex jam nut of MIL-S-1222H shall be substituted for the type IV semi-finished hexagon jam nut of MIL-B-857A. The type V hex slotted nut of MIL-S-1222H is interchangeable with the type VII finished hexagon slotted nut of MIL-B-857A. The type V hex slotted nut of MIL-S-1222H shall be substituted for the type VI regular semi-finished hexagon slotted nut of MIL-B-857A. The type V hex slotted nut of MIL-S-1222H has closer tolerances than the type VI regular semi-finished hexagon slotted nut of MIL-B-857A.

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6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Bolts
Hex cap screws
Nuts
Socket head cap screws

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 53GP-0109)

Review activities:
Navy - MC
DLA - IS

User activities:
Army – AR, AV, ME

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